



4th International Conference on Data Analytics and Insights : Submission (371) has been created.
1 message

Microsoft CMT <noreply@msr-cmt.org> Tue, Apr 21, 2026 at 10:11 AM
To: adurisaicharanreddy@gmail.com

Hello,

The following submission has been created.

Track Name: Wireless and Sensor Communication Systems

Paper ID: 371

Paper Title: A Scalable AI-Based System for Dynamic Traffic Signal Optimization and Real-Time Parking Space Detection

Abstract:
As cities grow rapidly and the number of cars on the road increases, a problem of traffic congestion and lack of parking spots has become one of the most critical issues in the contemporary cities. Conventional traffic control systems are usually based on the fixed or semi-adaptive control means and are not responsive to the real time situation. In this paper, a smart AI-based traffic and parking management system will be introduced, which combines machine learning and computer vision algorithms to overcome these challenges. The system is based on the Random Forest algorithm to process historical and live traffic data and predict congestion correctly and control traffic lights dynamically. Moreover, they use a computer vision module to track parking spots and see free parking lots in real-time, enabling users to find and book parking spots more effectively. The system proposed will enhance the general traffic, shorten the travel time, minimise congestion and give preference to emergency vehicles. Moreover, it offers a solution that is scalable and cost-effective and can be used in smart city transportation networks.
Index Terms–Predicting Traffic, Smart Parking, Machine Learning, Random Forest Algorithm, Intelligent Transportation Systems (ITS), Computer Vision, Real-Time Traffic Monitoring, Emergency Vehicle Prioritization, Smart Cities.

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Primary Subject Area: Artificial Intelligence,Machine Learning, Computer Vision, Intelligent Transportation Systems, Smart

Secondary Subject Areas: Not Entered

Submission Files:
Research_paper_154.pdf (2 Mb, Tue, 21 Apr 2026 04:41:11 GMT)

Submission Questions Response: Not Entered

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